

Content recommendations and ad targeting are the results of leveraging Spark's efficient iterative execution, which has been done by two of the world's leading companies. There is a good comparison between machine learning algorithms and general algorithms regarding speed. Machine learning algorithms (e.g., based on Spark logistic regression) run 100x more rapidly than previously implemented Hadoop-based algorithms, while other algorithms (collaborative filtering or other alternative multipliers) run 15x faster.

- **Real time streaming:** Spark offers the advantage of single semantics and full recovery of stateful operators. It also has the unique facility of providing you the same Spark APIs to process streams, including reusing the regular application code of Spark. It has an API for working with streams with low latency data analysis systems, and it can easily be extended further to process and analyse live data streams.
- **Quick design making:** Reducing latency and improving quality with efficiency are the advantages of Spark. Many companies use Big Data for recommendation systems, ad targeting or predictive analytics. Effectiveness increases by reducing decision latency. This leads to dramatic improvements in the company's ROI (return on investment). Spark deployment is ideal for ad targeting and improved video delivery over the Web.

Applications of Apache Spark

Apache Spark is currently used by many research projects at Berkeley related to machine learning, and it is also being studied at some Internet companies.

- **Interactive queries on data streams:** Quantified (a start-up) used Spark for the exploration of time series data. It was used to create a user friendly interactive interface. Quantified specialises in predictive analysis on time series data. The process contains three steps: to upload new data from external sources (e.g., Facebook or Twitter streams) at fixed periods, executing an entity extraction algorithm (for extracting the data), and in the end, it builds an in-memory table of each mentioned entity. With the help of Web applications, users can raise queries in a user friendly manner.
- **Spam detection on Twitter:** This is a research project at Berkeley. Apache Spark was used to investigate spam links in Twitter posts. A regression classifier for investigation purposes was developed and it was placed on top of Spark. With the help of Spark, the research team could develop an accurate and extremely fast classifier for spam detection.
- **Hive on Spark:** Spark offers full support to the HiveQL language. It can be run over Hive data warehouses as well. It enhances the capability of Hive by facilitating the loading of tables in-memory for much faster access. Apart from SQL support, it also provides machine learning functions like clustering and classification.
- **Spark streaming:** Spark streaming is a sub-project of

Spark. It extends the functionality of online processing by providing the MapReduce and filter functionality of Spark.

Setting up Spark on Amazon EC2

To set up a Spark cluster on Amazon EC2, you need to follow the simple steps listed below.

Pre-requisites: In order to access any service of the Amazon cloud, you first have to sign up with Amazon and then navigate to its EC2 service. After that you need to click on the 'Create key pair' tab (in the key pair section). You will be given the option of creating and downloading the key pair.

Downloading Spark: You can download Spark on your local machine by running the following command:

```
$ wget http://spark-project.org/files/spark-0.7.2-prebuilt-hadoop1.tgz
$ tar xvfz spark-0.7.2-prebuilt-hadoop1.tgz
```

The above command will create one EC2 directory, which contains the cluster set up script:

```
$ spark-0.7.2/ec2/spark-ec2 -k <keypair-name> -i <key-file> -s <num-slaves> launch <cluster-name>
```

Here *<keypair-name>* is the name of the EC2 key pair and *<key-file>* is the private key.

You can give the number of the slave that you want to launch in the *<num-slaves>* options, e.g., 1, 2, 3 or 4, etc.

You can give any suitable name to your cluster in the *<cluster-name>* option.

Launching a Spark cluster: Now, just log in to the master instance with the following command:

```
$ ./spark-ec2 -k key -i key.pem login <cluster-name>
```

For specific configurations, there is the */root/spark/conf/spark-env.sh* file on each machine, which lets you edit that file.

Terminating a cluster: If you want to terminate the cluster, run the following command:

```
$ ./spark-ec2 destroy <cluster-name>. 
```

References

- [1] http://docs.sigmoidanalytics.com/index.php/Running_Spark_on_EC2
- [2] <https://spark.incubator.apache.org/docs/latest/ec2-scripts.html>

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